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A Strategy to Achieve Sub-Parts-per-Million Mass Measurement Accuracy of *N*-Linked Glycans Using Infrared Matrix-Assisted Laser Desorption Electrospray Ionization

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ABSTRACT

Mass calibration techniques are vital in achieving high mass measurement accuracy (MMA) of large biomolecules. Variable ion populations that shift the axial frequencies due to space charge effects have been a significant challenge in achieving sub-parts-per-million (sub-ppm) MMA of glycans on a high-resolution accurate mass (HRAM) orbitrap instrument without the activation of automatic gain control. As the role of glycans is critical to our understanding of diverse biological processes, accurate identification of glycans using sub-ppm MMA is critical for biological interpretations. Hence, this study aims to achieve sub-ppm MMA of glycans by exploring the impact of different ion accumulation times, data collection modes, in addition to custom calibration strategies and external mass correction to optimize accurate mass measurements. Using infrared matrix-assisted laser desorption electrospray ionization (IR-MALDESI), direct analysis was performed on *N*-linked glycans cleaved from bovine fetuin in negative polarity, where 17 *N*-linked glycans were detected and annotated. Our results indicate the significance of implementing a custom calibration external lock mass and other techniques, including the effect of external mass correction in achieving sub-ppm MMA of large biomolecules. Implementing such approaches in mass spectrometry imaging (MSI) of biological tissue will enhance the confidence of glycan annotation and enable more accurate biological conclusions.

1 | Introduction

Mass spectrometry (MS) has been widely employed as a fundamental analytical technique for analyzing various biomolecules [1–3], and it has been implemented as an imaging technique in mass spectrometry imaging (MSI) platforms to detect and obtain spatial information of molecular species [4–6]. Over the years, the field of MS has experienced drastic improvement aimed at obtaining accurate mass-to-charge ratios (m/z) at sub-ppm MMA levels for both large and small biomolecules to improve the confidence in compound identification [7–9]. Although the calibration of mass spectrometers is routinely utilized to align instrument measurements to known reference ions from a calibration solution to ensure and maintain data reliability, reproducibility, precision, and accuracy, different calibration techniques and the choice of calibration solutions have been reported to influence the mass accuracy of molecules analyzed using electrospray ionization mass spectrometry (ESI-MS) [10, 11]. Such observations are further exacerbated in the field of glycomics due to the extensive structural heterogeneity derived from varying linkage types, branching, and different monosaccharide residues [12]. Glycans are biomolecules that play crucial roles in various cellular processes including cell–cell recognition, signaling, protein folding, and immune responses

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[13–16]. Their role in these biological processes has linked them to diverse life-threatening diseases including cancer, diabetes, Alzheimer's disease, and genetic disorders [17–19]. This poses the need for good calibration techniques and an optimized mass correction approach to achieve high MMA for accurate glycan characterization to minimize false identification and improve confidence in glycan annotations.

The advantages of calibrating mass spectrometers using ion mixtures that closely mimic the ionization behavior of the target analytes have been previously discussed [11, 20]. For instance, in protein analysis, the multiple charge states of intact proteins can serve as a useful tool for interpreting the spectra of other intact proteins [21, 22]. Although the use of different synthetic peptides as calibrant solutions has been reported [23–25], a calibrant solution containing varying lengths but a homogeneous composition of synthetic peptides called "PAS-cal" was customized using PASylation technology to suit calibration as intact charge envelope and as tryptic digest to address the challenges of complex and poorly resolved mass spectra [26]. Similarly, poly(thymidine) has been used for calibration in negative mode for DNA oligomers [27–29]. Moreover, the use of lipids in both negative and positive modes [30] and different lengths of sodium lauryl ether sulfate [31] and mixtures of fatty acids [32] as calibrants for lipid analysis in negative mode has been described. This mixture of compounds argued to offer closely spaced calibration points; however, the preparation and synthesizing process can be time consuming due to the need for careful optimization of concentration, *m/z* range coverage, and improved storage stability [33, 34]. Internal calibration by spiking a known standard into the sample has been shown to improve mass accuracy significantly [35, 36]; however, its implementation in MSI is limited. In MSI, the internal standard must be applied directly onto the tissue. This creates challenges in achieving the same ion abundance as the endogenous analyte within the tissue, which can eventually lead to ion suppression [37, 38].

As an alternative to previous reports, a strategy evaluating differing ion accumulation times and incorporating different data collection modes and calibration strategies would be valuable to the study of glycans. This study uses an Orbitrap Exploris 240 mass spectrometer, which is a high-resolution accurate mass (HRAM) spectrometer allowing for the separation of closely spaced isotopic peaks, determining elemental compositions, and identifying compounds [39]. A previous study by Kibbe et al. [39] demonstrated the effectiveness of HRAM in different data collection modes and the impact of external mass correction in achieving sub-ppm MMA without automatic gain control (AGC) for singly charged species with low *m/z* ratios ($<1000\,m/z$) in mouse liver. In this work, there is an extensive discussion of profile and centroid data collection modes. In centroid mode, sample molecules are ionized and separated by the mass analyzer based on their *m/z*. The raw signal collected by the detector is then processed by the instrument's built-in algorithms, which identify the maximum abundance within the peak and display a single centered point for each observed peak [39]. In profile mode, all peaks and their abundances are recorded as a series of data points along each peak. The selection of maximum abundance for centroid mode enhances accurate putative identification when compared to profile mode. Sub-ppm MMA has been achieved for small molecules [39]; therefore, this study seeks to extend this to multiply charged species, focusing on *N*-linked glycans with high *m/z* ratios typically ranging from 500 to 2000 to evaluate whether similar improvements in MMA can be achieved for more complex and large biomolecules. The complexity and higher *m/z* of glycans necessitate the investigation of space charge effects, which are common in trapping mass spectrometers such as Orbitrap, thereby emphasizing the need for ion accumulation time optimization [40]. Additionally, the use of custom calibration as opposed to the standard mass calibration to enable an external lock mass of interest, investigating data collection modes and the selection of the optimal mass correction ion for external mass correction rather than defaulting to the use of the most abundant ion. These strategies are vital in achieving sub-ppm MMA for *N*-linked glycans and are not limited to Orbitrap Exploris 240. Although this instrument was used, the custom calibration technique demonstrated is generally available on most high-resolution mass spectrometers, and when unavailable, calibrating with a solution that enables an external lock mass within the *m/z* range of the analyte would be beneficial. The strategy used to achieve sub-ppm MMA of *N*-linked glycan in this work improves glycan profiling and provides an optimized protocol for high mass accuracy detection of glycans.

2 | Experimental

2.1 | Sample Preparation

Bovine fetuin (Sigma-Aldrich, St. Louis, MO, USA) was analyzed following previously established protocols [41, 42]. 250 μg of bovine fetuin was loaded onto a 10 kDa molecular weight cutoff (MWCO) filter. To prepare the protein for enzymatic cleavage, 2 μL of 1 M dithiothreitol solution (Sigma-Aldrich, St. Louis, MO, USA) was added to denature the protein by reducing disulfide bonds and further diluted in 200 μL of 100 mM freshly prepared ammonium bicarbonate (Fisher Scientific Fair Lawn, NJ, USA) as digest buffer. The solution was lightly vortexed and incubated at 56°C for 30 min. The denatured proteins were then alkylated with 50 μL of 1 M iodoacetamide solution (Sigma-Aldrich, St. Louis, MO, USA) and incubated at 37°C for 60 min to prevent reformation of disulfide bonds. The alkylated, denatured proteins were washed three times with 100 μL digest buffer with three rounds of centrifugation for 20 min each time. The filters were transferred to a new centrifuge tube and 100 μL of optima LC/MS-grade water (Fisher Scientific, Waltham, MA, USA) was used to dissolve PNGase-F enzyme (Bulldog Bio, Portsmouth NH, USA). 2 μL of enzyme and 98 μL of digestion buffer were added to the filter to cleave *N*-linked glycans from the glycoprotein and incubated overnight at 37°C. Cleaved *N*-linked glycans were frozen at −80°C for 1 h, dried to completion using a speed vacuum cooler, and resuspended in 50 μL of optima LC/MS-grade water (Fisher Scientific, Waltham, MA, USA) to be directly analyzed by IR-MALDESI. The schematic workflow of this experiment is shown in Figure 1.

2.2 | IR-MALDESI Analysis

Mass and custom calibration procedures were performed with Pierce FlexMix (Thermo Fisher Scientific, Rockford, IL). For

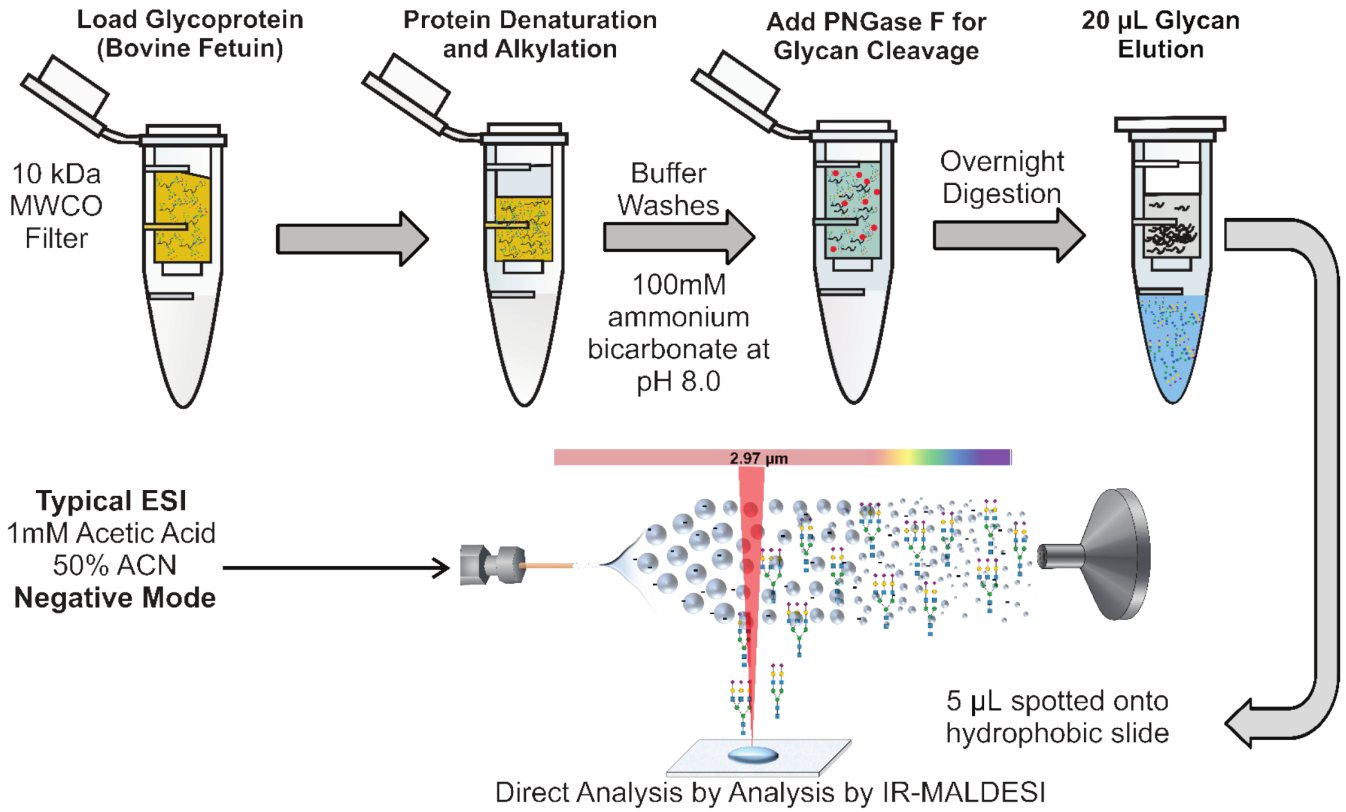


FIGURE 1 | Experimental workflow for direct analysis of *N*-linked glycans by IR-MALDESI.

custom calibration, the calibrants *m/z* outside the 500–2000 *m/z* range were unchecked on the Thermo Scientific Orbitrap Exploris 240 Tune software v4.2 SP4. Following calibration, 5 µL of the resuspended *N*-linked glycans were pipetted onto a Teflon microwell hydrophobic slide (Prosolia, Indianapolis, IN, USA) and subjected to IR-MALDESI coupled to an HRAM Orbitrap Exploris 240 mass spectrometer (Thermo Fisher Scientific, Bremen, Germany). A mid-IR laser (JGM Associates Inc., Burlington, MA, USA) firing 5 pulses per burst at a wavelength of 2970 nm produced 1.3 mJ of energy per burst to ablate neutral molecules, which were then ionized within the orthogonal stable electrospray plume. ESI was performed in negative mode with a voltage of 3.4 kV and a 1.5 µL/min flow rate applied to the electrospray solvent containing 50% acetonitrile and 1 mM acetic acid (Fischer Scientific, Nazareth, PA, USA). Resolving power (RP) was set 240000_{FWHM} at *m/z* 200, and the scan range was set to *m/z* 500–2000. AGC was turned off to enable a fixed injection time [43]. Data were collected in both centroid and profile modes.

2.3 | Data Analysis

All raw spectra collected were viewed in Thermo Scientific XCalibur v4.1.50. The *N*-linked glycan monoisotopic masses of each data collection mode were calculated, and the experimental masses were searched in GlycoMod [44]. The corresponding *N*-linked glycan structures were confirmed using GlyConnect, which is an *N*-linked glycan online database [44]. Raw spectra from both data collection modes were converted to .mzML format using MSConvert by Proteowizard [45]. .mzML was further converted to .imzML using imzMLConverter [46]. The theoretical mass of the selected *N*-linked glycans was used for single-point external mass correction on MSiReader v3.02 (MSI Software Solutions LLC, Raleigh, NC, USA) [46, 47]. The resulting MMA values were extracted to .xlsx files. Data were visualized using RStudio v2024.04.2 + 764 and Excel Microsoft 365 v2505 [39].

3 | Results and Discussion

The main objective of this study was to establish strategies for achieving sub-ppm MMA for *N*-linked glycans with potential applicability to other large biomolecules. In this study, we discuss the significance of the custom calibration approach, the effect of ion injection time on *N*-linked glycan abundance, detection sensitivity, MMA when analyzed in an ESI-like manner, the impact of data acquisition modes, and the advantages of external mass correction. The Orbitrap Exploris 240 was calibrated with FlexMix calibration solution using the custom calibration approach. This approach enables an external lock mass of *m/z*1033.98810, which is close to the *m/z* of *N*-linked glycans. This external lock mass corresponds to hexakis(2,2,3,3-tetrafluoropropoxy)phosphazene in FlexMix calibration solution. Data collected following custom calibration were used for evaluation and figure illustrations unless otherwise stated. To directly evaluate the effect of ion accumulation on spectral quality, the instrument's AGC was disabled, allowing ion injection time to serve as the primary determinant of the number of ions accumulated in the C-trap prior to analysis. A range of fixed injection times (15, 30, 45, 60, 75, 90, and 105 ms) were sequentially examined in profile acquisition mode. Although sensitivity and

signal abundance can be accessed in centroid mode, profile mode was specifically employed to accurately evaluate peak shape and resolution [48], thereby allowing a more comprehensive determination of the optimal injection time for *N*-linked glycans. The abundance of values corresponding to the highest *N*-linked glycan observed at *m/z* 958.3349 was used for demonstrating this phenomenon as this peak is expected to appear in all scans. The effect of injection time was further examined on the MMA of the same *N*-linked glycan. The same trend is expected for other *N*-linked glycans. The result is presented in Figure 2.

This result shows a consistent increase in ion abundance as injection time increases up to 90 ms. This reflects the significance of high *m/z* values of *N*-linked glycans on ion accumulation time. Due to *N*-linked glycans higher *m/z* values, their ions travel slowly, resulting in fewer ions reaching the C-trap and accumulating within the ion routing multipole before being ejected to the Orbitrap analyzer. Consequently, glycans of higher abundance were detected at short injection time, whereas low abundance glycans were not detected. A decline in signal abundance observed for 105 ms (result not shown) suggests potential C-trap saturation resulting from excessive ion accumulation. 90 ms is established to be the optimal ion accumulation time for glycan analysis, which makes it different from small molecules that typically require a shorter injection time of approximately 15 ms [39] and antibodies that require a longer injection time of 125 ms [43].

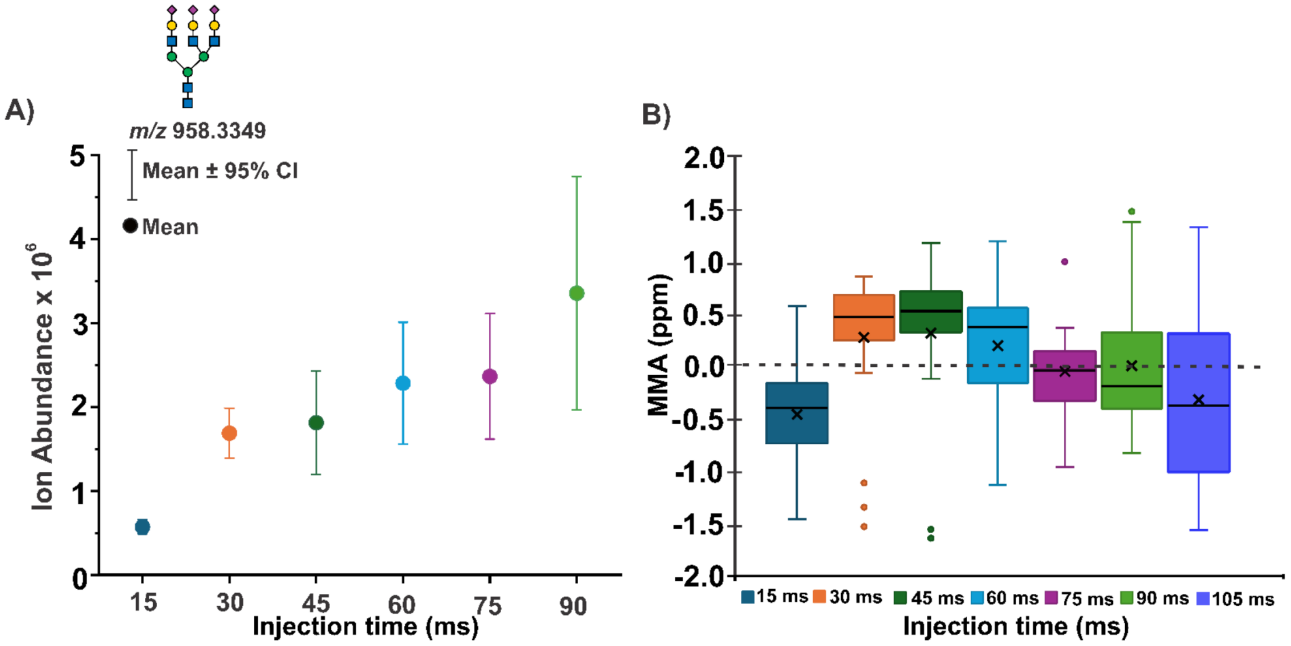


FIGURE 2 | (A) An interval plot showing the abundances exported across 25 scans for *m/z* 958.3349 against different injection times following custom calibration. The circle bulb represents the mean of the ion abundance. (B) Box plots showing the distribution of MMA for different ion injection time for *N*-linked glycan. The mean represented by the multiplication sign aligned with zero for the established optimal injection time and the widest variation in MMA distribution was observed for 105 ms. Data are presented as mean ± 95% confidence intervals.

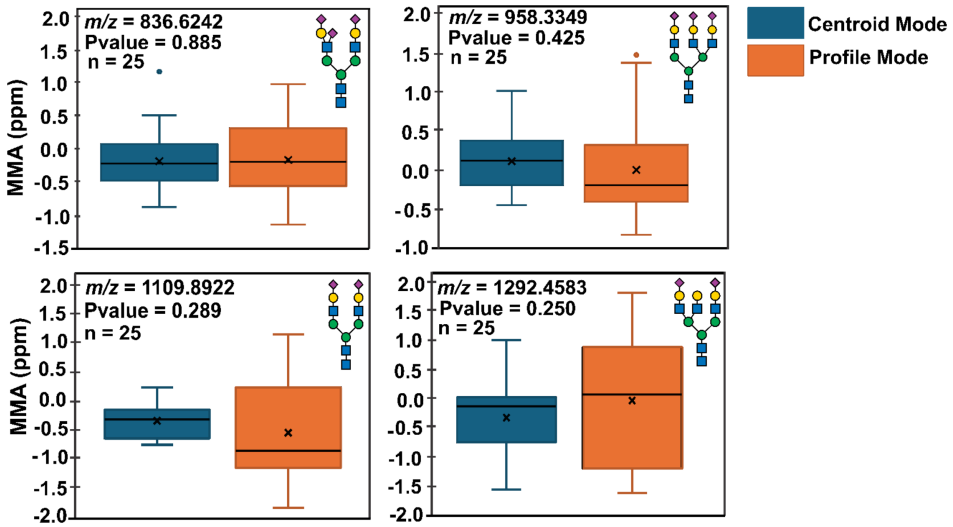


FIGURE 3 | Box plot comparing the two data acquisition modes in mass spectrometry of *N*-linked glycans. Centroid mode showed centralized distribution compared to profile mode.

The MMA for the injection times ranges from 0 to ±1.5, and this reflects the ability of the Orbitrap mass spectrometer to accurately determine the *m/z* of molecules of interest when calibration techniques are performed. After optimal injection time was established, complementary data were collected in centroid mode to compare data acquisition modes. Figure 3 illustrates the MMA distribution for four *N*-linked glycans selected across the mass spectrum. Centroid mode exhibits a centralized MMA distribution when compared to profile mode. This reflects the inbuilt data processing algorithm of both modes.

Previous studies on the analysis of bovine fetuin reported a lower number of *N*-linked glycans in negative mode. Using LC-ESI-IT-MS, 5 *N*-linked glycans were identified [49], whereas MALDI-MS detected 6, likely due to the labile nature of sialic acid residues on these glycans [50]. However, the soft ionization mechanism of IR-MALDESI [51] enabled the detection and analysis of intact sialylated *N*-linked glycans without chemical derivatization [52], resulting in the detection of 12 *N*-linked glycans, representing the highest recorded in the past 5 years. In the present study and as shown in Figure 4, 17 *N*-linked glycans were detected and identified on Glyconnect [44], and the structures were drawn on GlycoGlyph [53].

This increase can be explained using a longer injection time; whereas the original proof of concept used an injection time of 50 ms [42], the current analysis used 90 ms, which improved glycan detection by enhancing ion accumulation and ion

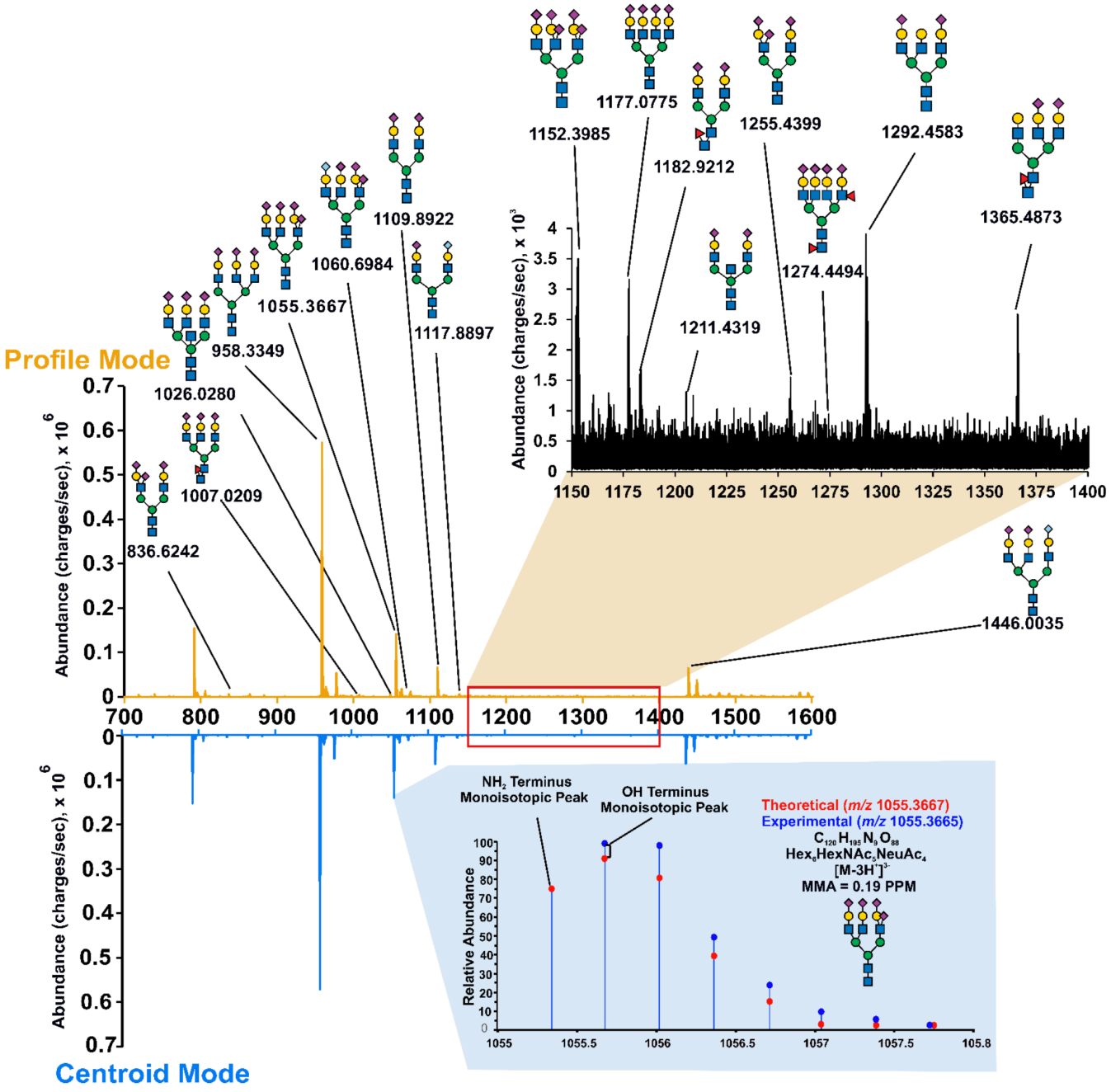


FIGURE 4 | Averaged full scan spectra of 25 scans for both mass spectrometry data collection modes: native profile (orange) and native centroid (blue) and spectral accuracy to show the isotopic distribution of Hex_6 $HexNAc_5$ $NeuAc_4$ (*m/z* 1055.3667).

abundance. The two spectra show that the same number of glycans were detected for both data acquisition modes, meaning no data lost, as presented in Figure 4. The *N*-linked glycans that are of low abundance were also visualized, and this is the same for both profile and centroid modes. To further demonstrate compositional structure conservation and mass spectral analysis of Hex_6 $HexNAc_5$ $NeuAc_4$ (m/z 1055.3667) was selected to illustrate the preservation of isotopic distribution, as seen in Figure 4.

The spectral accuracy reveals the presence of both the primary amine (NH_2) and hydroxyl (OH) termini on the cleaved *N*-linked glycan. Although PNGase F digestion usually produces *N*-linked glycans with a reducing end hydroxyl group, the use of ammonium bicarbonate buffer at an alkaline pH of 8.3 results in the glycan with an amine terminus [42, 54]. For an Orbitrap mass analyzer, resolving power (RP) decreases with the square root of the *m/z* [55]. The second isotopologue peak, which includes both the NH_2+1 (from one ^{13}C isotope in the amine group) and the OH monoisotopic mass, is observed as a single peak due to the reduced RP of the mass spectrometer at higher *m/z* values. The minimum RP required for accurate separation of these termini at this *m/z* is 163 547, which exceeds the observed RP of 103 705 and the theoretically calculated instrument RP of 104 477 at this *m/z*. Although the isotopic peaks are sufficiently resolved, this indicates glycans benefit from higher resolution MS to ensure accurate peak separation. The disparity observed in the abundance of each *N*-linked glycan is partially related to their actual concentration in the bovine fetuin but also attributed to their ionization behavior and the choice of charge states. As previously demonstrated in IR-MALDESI [42], a single *N*-linked glycan can exhibit multiple charge states with varying abundances [42].

Leveraging on the centralized distribution and less variability of the MMA values in centroid mode, we proceeded with centroid mode for data collection and post-data acquisition [39]. To investigate the effect of external lock mass, data collected after mass calibration and data collected after custom calibration were compared. Mass calibration using FlexMix gives a default external lock mass at *m/z* 318.97979, which is not within the *m/z* range for glycan analysis. Mass calibration has been employed in small molecule analysis and has proven to yield high MMA. However, for *N*-linked glycan analysis, custom calibration techniques were then employed to achieve an external lock mass of *m/z* 1033.98810, which is within the *m/z* range for routine *N*-linked glycan analysis. To demonstrate this, 12 *N*-linked glycans were strategically selected to cover the full *m/z* range of the detected glycan spectrum. A total of 279 counts were obtained for the data analysis. As shown in Figure 5, there is a significant improvement following custom calibration with a distinct shift of the MMA towards a sub-ppm MMA, where 81% of glycan measurements fell within sub-ppm level for custom calibration compared to 15% for routine mass calibration. This result suggests the use of custom calibration to achieve an external lock mass that is within the *m/z* range of the analyte of interest as a critical step in achieving sub-ppm MMA for *N*-linked glycans and other molecules with large mass including O-glycans, glycosphingolipid-derived glycans, and free oligosaccharides.

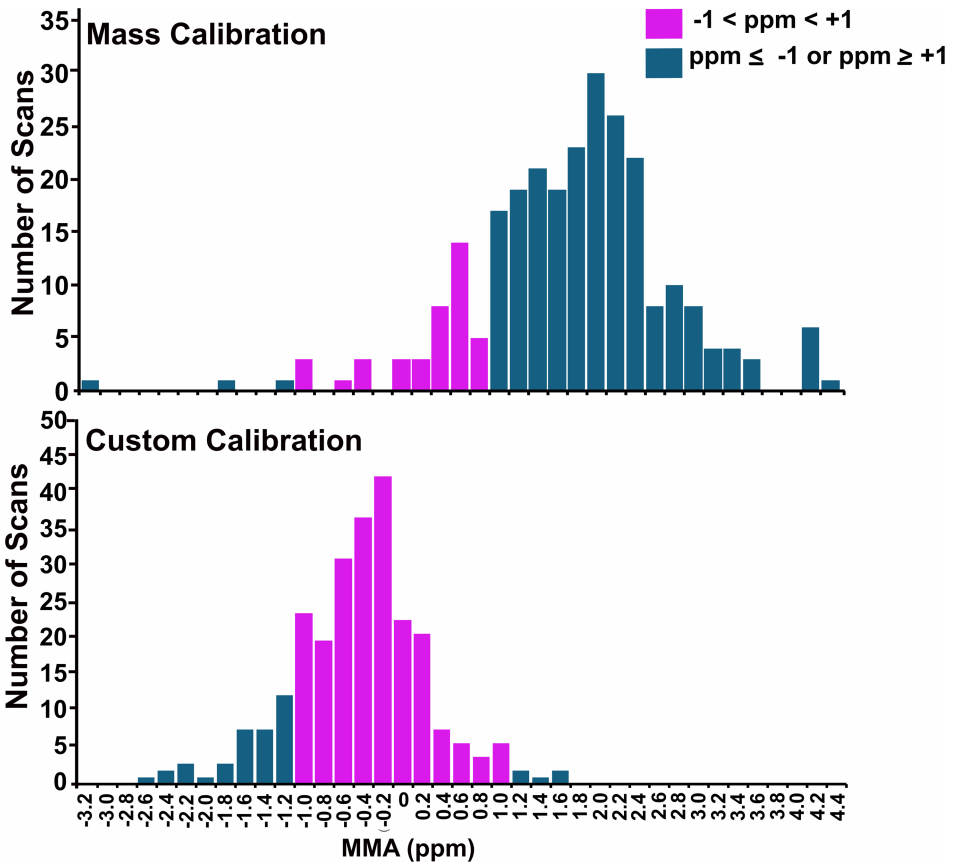


FIGURE 5 | Histograms of mass measurement accuracy for 12 selected N-linked glycans for 279 scans following mass calibration and custom calibration techniques. A shift of data distribution to sub-ppm MMA was observed for custom calibration.

The external lock mass serves as a reference within the calibration curve, allowing sub-ppm MMA to be achieved. However, a decline in MMA is expected as the analyte's *m/z* moves further away from the external lock mass. At a chosen external lock mass, the overall accuracy of the calibration curve is expressed as the root mean square (RMS); therefore, internal calibration known as the EASY IC ensures that the difference between the RMS obtained during calibration and that measured during analysis remains below 1 ppm. Consequently, sub-ppm MMA can be consistently achieved when the external lock mass is within the *m/z* range of the target analyte. In addition to the sialylated structures in Figure 4, achieving sub-ppm MMA further validates the preservation of sialic acid in the detected *N*-linked glycan (Figure 5).

To continue improving MMA, a single-point external mass correction was applied for data collected with custom calibration using the previously selected 12 *N*-linked glycans. Our aim is to identify an optimal reference *m/z* with MMA values that are evenly distributed, minimal deviation from zero, and the highest percentage within the sub-ppm range, rather than selecting the most intense peak for single-point external mass correction. To accomplish this, MMA values and experimental mass for each selected *N*-linked glycan were extracted using MSiReader. The theoretical *m/z* of the 12 *N*-linked glycans was used for single-point external mass correction. For calculating delta *m/z*, the spectrum was externally mass corrected sequentially with the selected 12 glycans, and their theoretical *m/z* was subtracted from the experimental masses of the selected glycans, including the glycan used for correction, which would give 0 MMA. The delta *m/z* values would either be positive or negative depending on the *m/z* used for mass correction and the experimental masses of other glycans. The MMA values were then plotted against the delta *m/z*, and the MMA values that fall within a sub-ppm were expressed as a percentage (Figure 6). Our findings show a similar MMA trend between *m/z* 1055.3667 and *m/z* 1109.8922. Both *m/z* 1055.3667 (94%) and 1109.8922 (90%) are closer to the external lock mass. However, *m/z* 1055.3667 was used as the optimal reference *m/z* for single-point external mass correction because it has a higher abundance in the spectrum when compared to *m/z* 1109.8922.

The MMA distribution of externally corrected glycans was further compared to the MMA distribution for custom

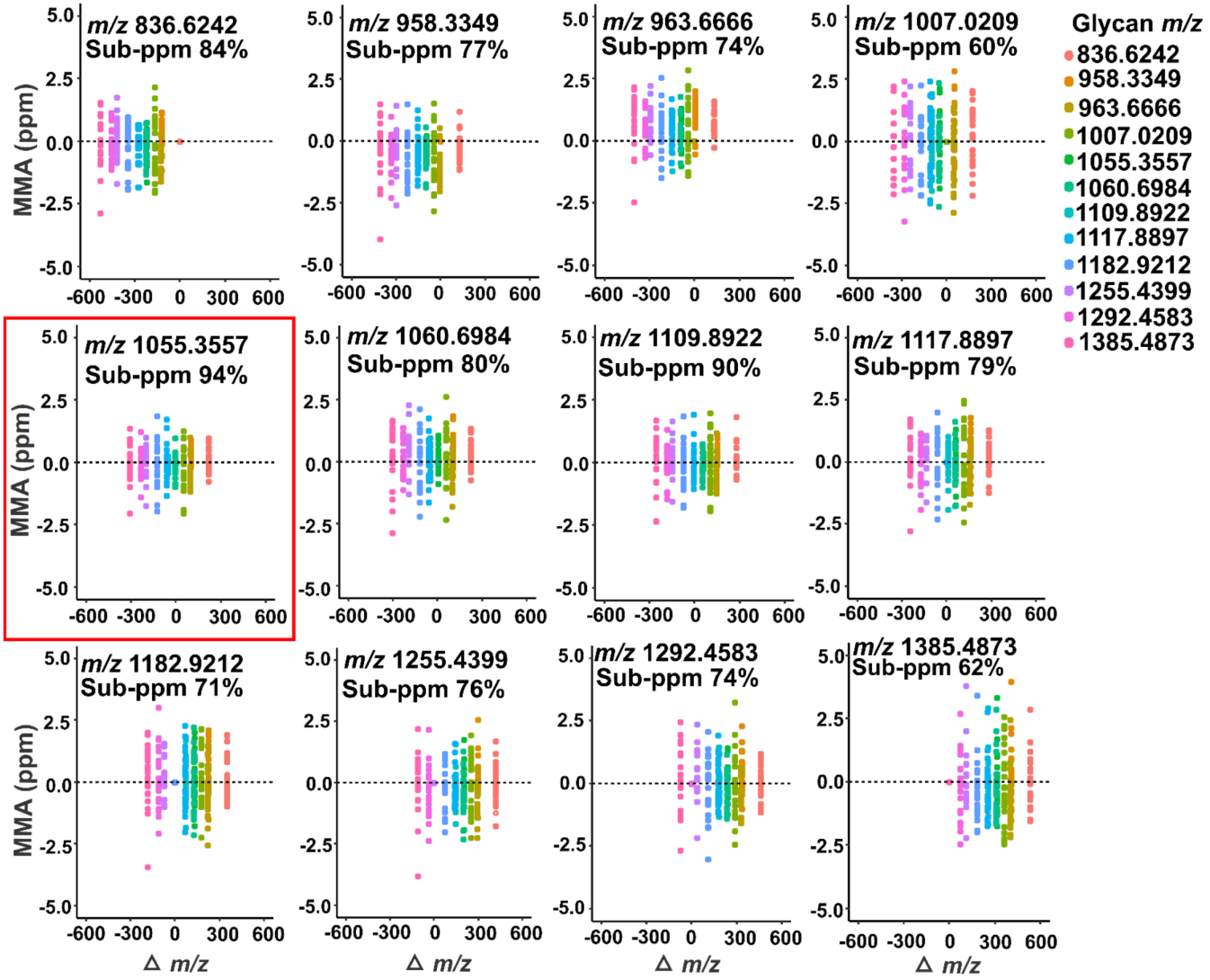


FIGURE 6 | Plot of MMA vs. Delta *m/z* illustrating the distribution of MMA values across the spectrum with the aim to identify *N*-linked glycan with minimal deviation from zero and the percentage of MMA values < 1 ppm.

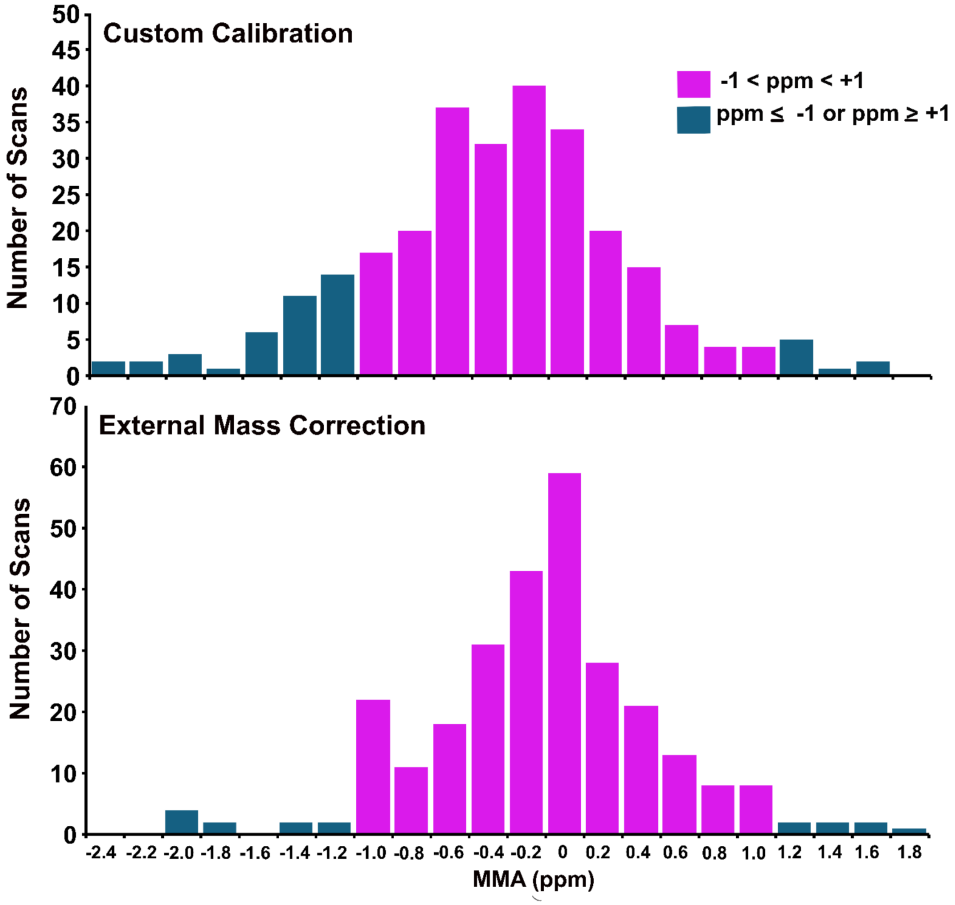


FIGURE 7 | Histogram showing MMA distribution for custom calibration and when it is externally mass corrected using the optimal reference *m/z* peak.

calibration. As shown in Figure 7, the use of a more intense *m/z* closer to the external lock mass would improve MMA and that external mass correction improves and allows a central distribution of MMA. A total of 279 counts were obtained before and after external mass correction. When externally mass corrected, 94% were recorded for the glycans that fall at sub-ppm levels compared to 81% without correction. The MMA values outside the sub-ppm range after external calibration were to be treated as outliers. Importantly, 100% of the *N*-linked glycans MMA values fall within the widely accepted 2.5 ppm.

4 | Conclusion

This study highlights practical methods for achieving a sub-ppm MMA of *N*-linked glycans by focusing on calibration techniques, ion accumulation time, data acquisition mode, and external mass correction approaches. As there is no loss of *N*-linked glycan peaks observed, collecting data in centroid mode demonstrated less data variability and a centralized distribution of MMA values. Injection time was observed to influence detection sensitivity, especially when the molecule of interest is a large molecule. The effect of external lock mass was also investigated through mass and custom calibration. By collecting data with the optimized injection time in centroid mode, employing the use of an external lock mass within the *m/z* of *N*-linked glycan, and a single-point external mass correction during data analysis post-acquisition, we achieved improved MMA in the analysis of *N*-linked glycans. The integration of these strategies for ESI-MS data will facilitate accurate profiling of *N*-linked glycans in MSI of biological samples.

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Conflicts of Interest

David C. Muddiman is a part owner of MSI Software Solutions LLC (Raleigh, NC). The other authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author.

References

1. M. Wuhrer, "Glycomics Using Mass Spectrometry," *Glycoconjugate Journal* 30, no. 1 (2013): 11–22, https://doi.org/10.1007/s10719-012-9376-3.

2. P. Lössl, M. van de Waterbeemd, and A. J. Heck, "The Diverse and Expanding Role of Mass Spectrometry in Structural and Molecular Biology," *EMBO Journal* 35, no. 24 (2016): 2634–2657, https://doi.org/10.15252/embj.201694818.

3. M. Caleb Bagley, K. P. Garrard, and D. C. Muddiman, "The Development and Application of Matrix Assisted Laser Desorption Electrospray Ionization: The Teenage Years," *Mass Spectrometry Reviews* 42, no. 1 (2023): 35–66, https://doi.org/10.1002/mas.21696.

4. J. H. Holbrook, G. E. Kemper, and A. B. Hummon, "Quantitative Mass Spectrometry Imaging: Therapeutics & Biomolecules," *Chemical Communications* 60, no. 16 (2024): 2137–2151, https://doi.org/10.1039/D3CC05988J.

5. S. Shimma, "Mass Spectrometry Imaging," *Mass Spectrometry* 11, no. 1 (2022): A0102, https://doi.org/10.5702/massspectrometry.A0102.

6. M. T. Bokhart and D. C. Muddiman, "Infrared Matrix-Assisted Laser Desorption Electrospray Ionization Mass Spectrometry Imaging Analysis of Biospecimens," *Analyst* 141, no. 18 (2016): 5236–5245, https://doi.org/10.1039/C6AN01189F.

7. A. Lommen, A. Gerssen, J. E. Oosterink, et al., "Ultra-Fast Searching Assists in Evaluating Sub-ppm Mass Accuracy Enhancement in U-HPLC/Orbitrap MS Data," *Metabolomics* 7, no. 1 (2011): 15–24, https://doi.org/10.1007/s11306-010-0230-y.

8. J. G. Stroh, C. J. Petucci, S. J. Brecker, N. Huang, and J. M. Lau, "Automated Sub-ppm Mass Accuracy on an ESI-TOF for Use With Drug Discovery Compound Libraries," *Journal of the American Society for Mass Spectrometry* 18, no. 9 (2007): 1612–1616, https://doi.org/10.1016/j.jasms.2007.06.001.

9. C. D. Wenger, G. C. McAlister, Q. Xia, and J. J. Coon, "Sub-Part-per-Million Precursor and Product Mass Accuracy for High-Throughput Proteomics on an Electron Transfer Dissociation-Enabled Orbitrap Mass Spectrometer," *Molecular & Cellular Proteomics* 9, no. 5 (2010): 754–763, https://doi.org/10.1074/mcp.M900541-MCP200.

10. Q. Duez, H. Metwally, and L. Konermann, "Electrospray Ionization of Polypropylene Glycol: Rayleigh-Charged Droplets, Competing Pathways, and Charge State-Dependent Conformations," *Analytical Chemistry* 90, no. 16 (2018): 9912–9920, https://doi.org/10.1021/acs.analchem.8b02115.

11. J. Romson and Å. Emmer, "Mass Calibration Options for Accurate Electrospray Ionization Mass Spectrometry," *International Journal of Mass Spectrometry* 467 (2021): 116619, https://doi.org/10.1016/j.ijms.2021.116619.

12. B. H. Li and X. S. Ye, "Recent Advances in Glycan Synthesis," *Current Opinion in Chemical Biology* 58 (2020): 20–27, https://doi.org/10.1016/j.cbpa.2020.04.009.

13. R. D. Cummings, "The Repertoire of Glycan Determinants in the Human Glycome," *Molecular BioSystems* 5, no. 10 (2009): 1087, https://doi.org/10.1039/b907931a.

14. A. Varki, "Biological Roles of Glycans," *Glycobiology* 27, no. 1 (2017): 3–49, https://doi.org/10.1093/glycob/cww086.

15. Y. van Kooyk and G. A. Rabinovich, "Protein-Glycan Interactions in the Control of Innate and Adaptive Immune Responses," *Nature Immunology* 9, no. 6 (2008): 593–601, https://doi.org/10.1038/ni.f.203.

16. E. N. Wallace, C. A. West, C. T. McDowell, et al., "An N-Glycome Tissue Atlas of 15 Human Normal and Cancer Tissue Types Determined by MALDI-Imaging Mass Spectrometry," *Scientific Reports* 14, no. 1 (2024): 489, https://doi.org/10.1038/s41598-023-50957-w.

17. W. S. York, R. Mazumder, R. Ranzinger, et al., "GlyGen: Computational and Informatics Resources for Glycoscience," *Glycobiology* 30, no. 2 (2020): 72–73, https://doi.org/10.1093/glycob/cwz080.

18. M. He, X. Zhou, and X. Wang, "Glycosylation: Mechanisms, Biological Functions and Clinical Implications," *Signal Transduction and Targeted Therapy* 9, no. 1 (2024): 194, https://doi.org/10.1038/s41392-024-01886-1.

19. T. V. Palomino and D. C. Muddiman, "Mass Spectrometry Imaging of N-Linked Glycans: Fundamentals and Recent Advances," *Mass Spectrometry Reviews* Published online June 27 (2024), https://doi.org/10.1002/mas.21895.

20. B. H. Clowers, E. D. Dodds, R. R. Seipert, and C. B. Lebrilla, "Dual Polarity Accurate Mass Calibration for Electrospray Ionization and Matrix-Assisted Laser Desorption/Ionization Mass Spectrometry Using Maltooligosaccharides," *Analytical Biochemistry* 381, no. 2 (2008): 205–213, https://doi.org/10.1016/j.ab.2008.06.041.

21. C. N. McEwen and R. S. Larsen, "Accurate Mass Measurement of Proteins Using Electrospray Ionization on a Magnetic Sector Instrument," *Rapid Communications in Mass Spectrometry* 6, no. 3 (1992): 173–178, https://doi.org/10.1002/rcm.1290060304.

22. H. S. Gadgil, G. D. Pipes, T. M. Dillon, M. J. Treuheit, and P. V. Bondarenko, "Improving Mass Accuracy of High Performance Liquid Chromatography/Electrospray Ionization Time-of-Flight Mass Spectrometry of Intact Antibodies," *Journal of the American Society for Mass Spectrometry* 17, no. 6 (2006): 867–872, https://doi.org/10.1016/j.jasms.2006.02.023.

23. J. Sztáray, A. Memboeuf, L. Drahos, and K. Vékey, "Leucine Enkephalin—A Mass Spectrometry Standard," *Mass Spectrometry Reviews* 30, no. 2 (2011): 298–320, https://doi.org/10.1002/mas.20279.

24. C. Eckers, J. C. Wolff, N. J. Haskins, A. B. Sage, K. Giles, and R. Bateman, "Accurate Mass Liquid Chromatography/Mass Spectrometry on Orthogonal Acceleration Time-of-Flight Mass Analyzers Using Switching Between Separate Sample and Reference Sprays. 1. Proof of Concept," *Analytical Chemistry* 72, no. 16 (2000): 3683–3688, https://doi.org/10.1021/ac000448i.

25. A. Staes, J. Vandenbussche, H. Demol, et al., "Asn_3, a Reliable, Robust, and Universal Lock Mass for Improved Accuracy in LC–MS and LC–MS/MS," *Analytical Chemistry* 85, no. 22 (2013): 11054–11060, https://doi.org/10.1021/ac4027093.

26. J. Breibeck, A. Serafin, A. Reichert, S. Maier, B. Küster, and A. Skerra, "PAS-Cal: A Generic Recombinant Peptide Calibration Standard for Mass Spectrometry," *Journal of the American Society for Mass Spectrometry* 25, no. 8 (2014): 1489–1497, https://doi.org/10.1007/s13361-014-0902-3.

27. J. M. Koomen, W. K. Russell, S. E. Tichy, and D. H. Russell, "Accurate Mass Measurement of DNA Oligonucleotide Ions Using High-Resolution Time-of-Flight Mass Spectrometry," *Journal of Mass Spectrometry* 37, no. 4 (2002): 357–371, https://doi.org/10.1002/jms.312.

28. A. P. Null and D. C. Muddiman, "Determination of a Correction to Improve Mass Measurement Accuracy of Isotopically Unresolved Polymerase Chain Reaction Amplicons by Electrospray Ionization Fourier Transform Ion Cyclotron Resonance Mass Spectrometry," *Rapid Communications in Mass Spectrometry* 17, no. 15 (2003): 1714–1722, https://doi.org/10.1002/rcm.1111.

29. H. Oberacher, H. Niederstätter, and W. Parson, "Characterization of Synthetic Nucleic Acids by Electrospray Ionization Quadrupole Time-of-Flight Mass Spectrometry," *Journal of Mass Spectrometry* 40, no. 7 (2005): 932–945, https://doi.org/10.1002/jms.870.

30. K. Ekroos and A. Shevchenko, "Simple Two-Point Calibration of Hybrid Quadrupole Time-of-Flight Instruments Using a Synthetic Lipid Standard," *Rapid Communications in Mass Spectrometry* 16, no. 12 (2002): 1254–1255, https://doi.org/10.1002/rcm.705.

31. R. Kostiainen, J. Tuominen, L. Luukkanen, J. Taskinen, and B. N. Green, "Accurate Mass Measurements of Some Glucuronide Derivatives by Electrospray Low Resolution Quadrupole Mass Spectrometry," *Rapid Communications in Mass Spectrometry* 11, no. 3 (1997): 283–285,

https://doi.org/10.1002/(SICI)1097-0231(19970215)11:3<283::AID-RCM792>3.0.CO;2-M.

32. R. L. Sleighter, G. A. McKee, Z. Liu, and P. G. Hatcher, "Naturally Present Fatty Acids as Internal Calibrants for Fourier Transform Mass Spectra of Dissolved Organic Matter," *Limnology and Oceanography: Methods* 6, no. 6 (2008): 246–253, https://doi.org/10.4319/lom.2008.6.246.

33. J. Zaia, R. S. Annan, and K. Biemann, "The Correct Molecular Weight of Myoglobin, a Common Calibrant for Mass Spectrometry," *Rapid Communications in Mass Spectrometry* 6, no. 1 (1992): 32–36, https://doi.org/10.1002/rcm.1290060108.

34. M. C. Lai and E. M. Topp, "Solid-State Chemical Stability of Proteins and Peptides," *Journal of Pharmaceutical Sciences* 88, no. 5 (1999): 489–500, https://doi.org/10.1021/js980374e.

35. D. K. Williams, A. M. Hawkridge, and D. C. Muddiman, "Sub Parts-per-Million Mass Measurement Accuracy of Intact Proteins and Product Ions Achieved Using a Dual Electrospray Ionization Quadrupole Fourier Transform Ion Cyclotron Resonance Mass Spectrometer," *Journal of the American Society for Mass Spectrometry* 18, no. 1 (2007): 1–7, https://doi.org/10.1016/j.jasms.2006.08.014.

36. W. L. Cheng, C. Markus, C. Y. Lim, R. Z. Tan, S. K. Sethi, and T. P. Loh, "Calibration Practices in Clinical Mass Spectrometry: Review and Recommendations," *Annals of Laboratory Medicine* 43, no. 1 (2023): 5–18, https://doi.org/10.3343/alm.2023.43.1.5.

37. Y. Li, N. Zhang, Y. Zhou, et al., "Induced Dual-Nanospray: A Novel Internal Calibration Method for Convenient and Accurate Mass Measurement," *Journal of the American Society for Mass Spectrometry* 24, no. 9 (2013): 1446–1449, https://doi.org/10.1007/s13361-013-0670-5.

38. D. A. Pirman, A. Kiss, R. M. A. Heeren, and R. A. Yost, "Identifying Tissue-Specific Signal Variation in MALDI Mass Spectrometric Imaging by Use of an Internal Standard," *Analytical Chemistry* 85, no. 2 (2013): 1090–1096, https://doi.org/10.1021/ac3029618.

39. R. R. Kibbe and D. C. Muddiman, "Achieving Sub-Parts-per-Million Mass Measurement Accuracy on an Orbitrap Mass Spectrometry Imaging Platform Without Automatic Gain Control," *Journal of the American Society for Mass Spectrometry* 34, no. 6 (2023): 1015–1023, https://doi.org/10.1021/jasms.3c00004.

40. U. M. Hohenester, P. Barbier Saint-Hilaire, F. Fenaille, and R. B. Cole, "Investigation of Space Charge Effects and Ion Trapping Capacity on Direct Introduction Ultra-High-Resolution Mass Spectrometry Workflows for Metabolomics," *Journal of Mass Spectrometry* 55, no. 10 (2020): e4613, https://doi.org/10.1002/jms.4613.

41. E. S. Hecht, J. P. McCord, and D. C. Muddiman, "A Quantitative Glycomics and Proteomics Combined Purification Strategy," *Journal of Visualized Experiments*, no. 109 (2016): 53735, https://doi.org/10.3791/53735.

42. C. L. Pace and D. C. Muddiman, "Direct Analysis of Native N-Linked Glycans by IR-MALDESI," *Journal of the American Society for Mass Spectrometry* 31, no. 8 (2020): 1759–1762, https://doi.org/10.1021/jasms.0c00176.

43. K. T. Knizner, M. C. Bagley, K. P. Garrard, et al., "Optimized C-Trap Timing of an Orbitrap 240 Mass Spectrometer for High-Throughput Screening and Native MS by IR-MALDESI," *Journal of the American Society for Mass Spectrometry* 33, no. 2 (2022): 328–334, https://doi.org/10.1021/jasms.1c00319.

44. J. Mariethoz, D. Alocci, A. Gastaldello, et al., "Glycomics@ExPASy: Bridging the Gap," *Molecular and Cellular Proteomics* 17, no. 11 (2018): 2164–2176, https://doi.org/10.1074/mcp.RA118.000799.

45. D. Kessner, M. Chambers, R. Burke, D. Agus, and P. Mallick, "ProteoWizard: Open Source Software for Rapid Proteomics Tools Development," *Bioinformatics* 24, no. 21 (2008): 2534–2536, https://doi.org/10.1093/bioinformatics/btn323.

46. M. T. Bokhart, M. Nazari, K. P. Garrard, and D. C. Muddiman, "MSiReader v1.0: Evolving Open-Source Mass Spectrometry Imaging Software for Targeted and Untargeted Analyses," *Journal of the American Society for Mass Spectrometry* 29, no. 1 (2018): 8–16, https://doi.org/10.1007/s13361-017-1809-6.

47. G. Robichaud, K. P. Garrard, J. A. Barry, and D. C. Muddiman, "MSiReader: An Open-Source Interface to View and Analyze High Resolving Power MS Imaging Files on Matlab Platform," *Journal of the American Society for Mass Spectrometry* 24, no. 5 (2013): 718–721, https://doi.org/10.1007/s13361-013-0607-z.

48. M. Reuschenbach, L. L. Hohrenk-Danzouma, T. C. Schmidt, and G. Renner, "Development of a Scoring Parameter to Characterize Data Quality of Centroids in High-Resolution Mass Spectra," *Analytical and Bioanalytical Chemistry* 414, no. 22 (2022): 6635–6645, https://doi.org/10.1007/s00216-022-04224-y.

49. M. Thaysen-Andersen, M. R. Larsen, N. H. Packer, and G. Palmisano, "Structural Analysis of Glycoprotein Sialylation – Part I: Pre-LC-MS Analytical Strategies," *RSC Advances* 3, no. 45 (2013): 22683, https://doi.org/10.1039/c3ra42960a.

50. M. H. J. Selman, M. Hoffmann, G. Zauner, et al., "MALDI-TOF-MS Analysis of Sialylated Glycans and Glycopeptides Using 4-Chloro-α-Cyanocinnamic Acid Matrix," *Proteomics* 12, no. 9 (2012): 1337–1348, https://doi.org/10.1002/pmic.201100498.

51. A. Tu and D. C. Muddiman, "Internal Energy Deposition in Infrared Matrix-Assisted Laser Desorption Electrospray Ionization With and Without the Use of Ice as a Matrix," *Journal of the American Society for Mass Spectrometry* 30, no. 11 (2019): 2380–2391, https://doi.org/10.1007/s13361-019-02323-2.

52. T. V. Palomino and D. C. Muddiman, "Predicting Sialic Acid Content of N -Linked Glycans Using the Isotopic Pattern of Chlorine," *Journal of the American Society for Mass Spectrometry* 34, no. 7 (2023): 1392–1399, https://doi.org/10.1021/jasms.3c00100.

53. A. Y. Mehta and R. D. Cummings, "GlycoGlyph: A Glycan Visualizing, Drawing and Naming Application," *Bioinformatics* 36, no. 11 (2020): 3613–3614, https://doi.org/10.1093/bioinformatics/btaa190.

54. J. Muzikar, Y. Mechref, Y. Huang, and M. V. Novotny, "Enhanced Post-Source Decay and Cross-Ring Fragmentation of Oligosaccharides Facilitated by Conversion to Amino Derivatives," *Rapid Communications in Mass Spectrometry* 18, no. 13 (2004): 1513–1518, https://doi.org/10.1002/rcm.1506.

55. A. G. Marshall, G. T. Blakney, T. Chen, et al., "Mass Resolution and Mass Accuracy: How Much Is Enough?," *Mass Spectrometry* 2, no. Special_Issue (2013): S0009, https://doi.org/10.5702/massspectrometry.S0009.